TS-14
AI-Powered Complaint Classification & Resolution Recommendation Engine

Dev IT Limited

PROBLEM STATEMENT:

In the wellness business, customer complaints are received through multiple channels such as call centres, emails, and direct communication. These complaints are manually reviewed and categorized (e.g., product issues, packaging issues, trade inquiries), leading to delays, inconsistent tagging, and missed service-level agreements (SLAs).

There is no intelligent mechanism to prioritize critical complaints or suggest appropriate resolution steps. As a result, response time increases and customer satisfaction declines.

A system is needed that can automatically classify complaints, assign priority based on urgency, and recommend actionable resolution steps in real-time—ensuring faster, consistent, and efficient customer support operations.

WINNING LOGIC:

- **Intelligent complaint classification:** System must classify complaints using meaningful logic (combination of keyword analysis and AI/ML models), not static rule-based mapping
- **Dynamic priority tagging:** Priority levels (High/Medium/Low) should be assigned based on complaint content, urgency, and impact rather than manual input
- **Actionable resolution recommendations:** System must suggest specific next steps (e.g., replacement, escalation, follow-up) instead of generic responses
- **Real-time processing:** Classification, priority tagging, and recommendation must be generated within seconds of complaint submission
- **Comprehensive dashboard visibility:** Dashboard should provide real-time insights into complaint categories, priorities, and workload distribution
- **Lifecycle tracking**: System must track complaints from submission to resolution, including status updates and SLA compliance

USE CASES

Customer Support Executive	Submit complaints via text/email/call summary; view auto-classified complaint category and assigned priority; follow system-recommended resolution steps; update complaint status after action
Quality Assurance Team	Analyze categorized complaints for accuracy; monitor trends in complaint types; identify

	recurring issues; ensure consistency in classification and resolution quality
Operations Manager	Monitor real-time dashboards showing complaint distribution and priorities; track SLA compliance and resolution time

FUNCTIONALITIES TO BE ACHIEVED:

1. Complaint input system (text/email/call summaries)
2. Automatic complaint classification (Product / Packaging / Trade)
3. Priority tagging (High / Medium / Low)
4. AI-based resolution recommendation engine
5. SLA tracking for each complaint
6. Dashboard with complaint category and priority insights
7. Complaint history and status tracking
8. Reporting and export functionality (PDF/CSV)

DATA SOURCE

College event and project submission forms; GTU affiliated college registry (400+ institutions); student participation records.

EVALUATION CRITERIA

All teams are evaluated on the same universal rubric by the judging panel.

CRITERIA	WEIGHT
Innovation & Creativity	**25 marks**
Technical Complexity & Implementation	**25 marks**
Problem Relevance & Impact	**20 marks**
Demo & Presentation Quality	**15 marks**
Code Quality & Documentation	**10 marks**
Checkpoint Progress Score	**5 marks**
TOTAL	**100 marks**